

CPCS 214 - Final Exam

	Part I (SO 1)					Sub-total	Part II				Sub total	Total
	1	2	3	4	5		6	7	8	9		
Pts	/12	/6	/10	/12	/10		/10	/12	/18	/10		
CLO	2	5	8	10	11		3	4	7	9		

Answer all questions. Total points : 100

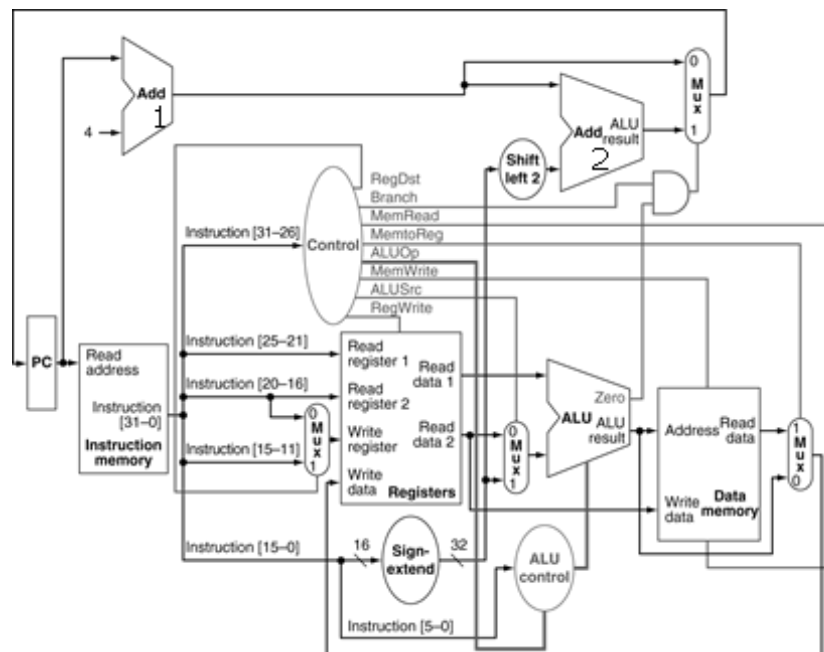
Part I – 50 points

1. Compile the following program to MIPS assembly. Use the variable-register mapping: n: \$t0 and base register of A: \$s0.
(CLO-2, SO-1) (12 pts)

```
int x = 1;
while (x<7)
{
    A[x+1] = A[x] + 4;
    x++;
}
```

2. Convert the number: (-87.75), to the standard IEEE single-precision format. (CLO-5, SO-1) (6 points)

3. The single-cycle 32-bit architecture is shown in the next figure. (CLO-8, SO-1) (10 pts)



Assume that the following instruction is executed on this architecture.

lw \$s3, 16(\$s1)

a. How many bits are used to store the constant 16 in the instruction above? Is it required to convert this to use additional bits? If so, specify the functional units in the figure which are needed to achieve this. What is the role of “Shift left 2” in the execution of this instruction?

(4 points)

b. What is the use of ALU1 and ALU2 for this instruction? (3 points)

c. Is Zero of the main ALU used in this instruction? If not, which output of the main ALU is used in this instruction?

d. What are the values of these control elements for this instruction? (2 points)

MemtoReg	MemRead	RegDest	ALUSrc

4. The following assembly program is executed on a 5-stages (IF, ID, EXE, MEM, WB) pipelined architecture with two separate instruction and data memories: (CLO-10, SO-1) (12 pts)

```
1: add $s1, $s2, $s3
2: lw $t1, 8($s2)
3: add $t3, $s1, $t2
4: sw $t1, 12($s2)
```

a. Draw a pipelined diagram (using blocks) with all necessary stalls and forwarding paths.

b. What is the total number of clock cycles needed to execute this program?

5. a. In the following MIPS instruction sequence, which type of hazard is produced in a pipelined system? and at which instruction? (CLO-11, SO-1) (4 points)

```
lw $s2, 10($s1)
sw $s3, 20($s1)
bne $s0, $zero, Exit
add $s4, $s3, $s6
...
Exit: ...
```

b. In general, what causes this type of hazard in a pipelined data path, and how can be eliminated? (4 points)

c. Propose a solution to avoid this type of hazard and explain how it can be done. (2 points)

Part II – 50 points

6. What are the contents of \$t0 and \$t2 after the four indicated instructions? (CLO-3, SO-1) (10 pts)

```
addi $t0, $zero, 8           # instruction1, $t0 =      , $t2 =
addi $t2, $zero, -4          # instruction2, $t0 =      , $t2 =
sll  $t2, $t0, 2             # instruction3, $t0 =      , $t2 =
bne  $t2, $t0, ELSE
j    DONE
```

```
ELSE: addi $t2, $t2, -3
```

```
# instruction4, $t0 =      , $t2 =
```

```
DONE:
```

7. Write a method(procedure) that implements the min operation using MIPS code. The min method takes two numbers as input parameters and returns the smaller of the two numbers. If the numbers are equal then either of the two numbers could be returned. Do not use pseudo-instructions. (CLO-4, SO-1) (12 pts)

